

Simple harmonic oscillator

Xiao Mingwen

Nanjing University

Hamiltonian

$$H = \frac{\mathbf{p}^2}{2m} + \frac{1}{2}kx^2,$$

$$\omega = \sqrt{\frac{k}{m}},$$

$$H = \frac{\mathbf{p}^2}{2m} + \frac{1}{2}m\omega^2x^2,$$

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2.$$

Dimensionless

$$H = \frac{1}{2}\hbar\omega \left[-\frac{\hbar}{m\omega} \frac{d^2}{dx^2} + \frac{m\omega}{\hbar} x^2 \right],$$

$$\xi = \alpha x = \sqrt{\frac{m\omega}{\hbar}} x,$$

$$H = \frac{1}{2}\hbar\omega \left[-\frac{d^2}{d\xi^2} + \xi^2 \right].$$

Schrödinger equation

$$H\psi(x) = E\psi(x),$$

$$\psi(x) \rightarrow 0, \quad x \rightarrow \pm\infty.$$

$$\frac{1}{2}\hbar\omega \left[-\frac{d^2}{d\xi^2} \psi + \xi^2 \psi \right] = E\psi,$$

$$\psi(\xi) \rightarrow 0, \quad \xi \rightarrow \pm\infty.$$

Schrödinger equation

$$\frac{1}{2}\hbar\omega \left[-\frac{d^2}{d\xi^2} \psi + \xi^2 \psi \right] = E \psi,$$

$$\lambda = \frac{E}{\frac{1}{2}\hbar\omega},$$

$$-\frac{d^2}{d\xi^2} \psi + \xi^2 \psi = \lambda \psi,$$

$$\frac{d^2}{d\xi^2} \psi + (\lambda - \xi^2) \psi = 0.$$

$$\frac{d^2}{d\xi^2}\psi + (\lambda - \xi^2)\psi = 0.$$

$$x \rightarrow \pm\infty,$$

$$\frac{d^2}{d\xi^2}\psi - \xi^2\psi = 0.$$

$$\psi \propto \exp\left(\pm\frac{1}{2}\xi^2\right).$$

$$\exp\left(\frac{1}{2}\xi^2\right) \rightarrow +\infty,$$

$$\psi \propto \exp\left(-\frac{1}{2}\xi^2\right).$$

Transformation

$$\psi(\xi) = u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right),$$

$$\frac{d\psi}{d\xi} = \frac{du}{d\xi} \cdot \exp\left(-\frac{1}{2}\xi^2\right) - \xi u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right),$$

$$\begin{aligned}\frac{d^2}{d\xi^2} \psi &= \frac{d^2 u}{d\xi^2} \cdot \exp\left(-\frac{1}{2}\xi^2\right) - 2\xi \frac{du}{d\xi} \cdot \exp\left(-\frac{1}{2}\xi^2\right) \\ &\quad - u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right) + \xi^2 u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right),\end{aligned}$$

Transformation

$$\begin{aligned}& \frac{d^2}{d\xi^2} \psi + (\lambda - \xi^2) \psi \\&= \frac{d^2 u}{d\xi^2} \cdot \exp\left(-\frac{1}{2}\xi^2\right) - 2\xi \frac{du}{d\xi} \cdot \exp\left(-\frac{1}{2}\xi^2\right) \\& \quad - u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right) + \xi^2 u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right) \\& \quad + (\lambda - \xi^2) u(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right) \\&= \left[\frac{d^2 u}{d\xi^2} - 2\xi \frac{du}{d\xi} + (\lambda - 1)u \right] \cdot \exp\left(-\frac{1}{2}\xi^2\right),\end{aligned}$$

Transformation

$$\frac{d^2}{d\xi^2} \psi + (\lambda - \xi^2) \psi = \left[\frac{d^2 u}{d\xi^2} - 2\xi \frac{du}{d\xi} + (\lambda - 1)u \right] \cdot \exp\left(-\frac{1}{2}\xi^2\right).$$

$$\frac{d^2}{d\xi^2} \psi + (\lambda - \xi^2) \psi = 0,$$

$$\frac{d^2 u}{d\xi^2} - 2\xi \frac{du}{d\xi} + (\lambda - 1)u = 0.$$

Hermite Equation

$$u'' - 2zu' + (\lambda - 1)u = 0.$$

Solution to Hermite equation

$$u'' - 2zu' + (\lambda - 1)u = 0.$$

$$u(z) = \sum_{k=0}^{+\infty} C_k z^k,$$

$$u' = \sum_{k=0}^{+\infty} k C_k z^{k-1},$$

$$u'' = \sum_{k=0}^{+\infty} k(k-1) C_k z^{k-2} = \sum_{k=0}^{+\infty} (k+2)(k+1) C_{k+2} z^k$$

Solution to Hermite equation

$$u'' - 2zu' + (\lambda - 1)u$$

$$= \sum_{k=0}^{+\infty} (k+2)(k+1) C_{k+2} z^k - \sum_{k=0}^{+\infty} 2k C_k z^k + \sum_{k=0}^{+\infty} (\lambda - 1) C_k z^k$$

$$= \sum_{k=0}^{+\infty} \{ (k+2)(k+1) C_{k+2} - [2k - (\lambda - 1)] C_k \} z^k = 0,$$

$$C_{k+2} = \frac{2k - (\lambda - 1)}{(k+2)(k+1)} C_k.$$

Solution to Hermite equation

$$u'' - 2zu' + (\lambda - 1)u = 0.$$

$$u(z) = \sum_{k=0}^{+\infty} C_k z^k,$$

$$C_{k+2} = \frac{2k - (\lambda - 1)}{(k + 2)(k + 1)} C_k.$$

$$u_1(z) = C_0 + C_2 z^2 + C_4 z^4 + \cdots,$$

$$u_2(z) = C_1 z + C_3 z^3 + C_5 z^5 + \cdots.$$

$$\lambda \neq 2n + 1,$$

$$\lim_{k \rightarrow +\infty} \frac{C_{k+2}}{C_k} = 0 \Rightarrow \rho = +\infty.$$

$$\frac{C_{k+2}}{C_k} \rightarrow \frac{1}{k},$$

$$\exp(z^2) = \sum_{k=0}^{+\infty} \frac{1}{k!} z^{2m}.$$

$$u_1(z) \propto \exp(z^2) \rightarrow +\infty, \quad z \rightarrow \infty,$$

$$u_2(z) \propto z \exp(z^2) \rightarrow \infty, \quad z \rightarrow \infty.$$

$$\lambda \neq 2n + 1,$$

$$\psi_1(\xi) = u_1(\xi) \exp\left(-\frac{1}{2}\xi^2\right) \propto \exp\left(\frac{1}{2}\xi^2\right) \rightarrow +\infty, \quad \xi \rightarrow \pm\infty,$$

$$\psi_2(\xi) = u_2(\xi) \exp\left(-\frac{1}{2}\xi^2\right) \propto \xi \exp\left(\frac{1}{2}\xi^2\right) \rightarrow \pm\infty, \quad \xi \rightarrow \pm\infty.$$

Alert

$\lambda \neq 2n + 1$, $\psi(\xi)$ can not satisfy the boundary condition!

$$\lambda = 2n + 1, \quad n = 0, 1, 2, \dots,$$

$$C_{k+2} = \frac{2k - (\lambda - 1)}{(k + 2)(k + 1)} C_k = -\frac{2(n - k)}{(k + 2)(k + 1)} C_k.$$

$$C_{n+2} = C_{n+4} = C_{n+6} = \dots = 0,$$

$$u_1(z) = C_0 + C_2 z^2 + \dots + C_n z^n, \quad n = \text{even number},$$

$$u_2(z) = C_1 z + C_3 z^3 + \dots + C_n z^n, \quad n = \text{odd number}.$$

$$\psi_1(\xi) = u_1(\xi) \exp\left(-\frac{1}{2}\xi^2\right) \propto \xi^n \exp\left(-\frac{1}{2}\xi^2\right) \rightarrow 0, \quad \xi \rightarrow \pm\infty,$$

$$\psi_2(\xi) = u_2(\xi) \exp\left(-\frac{1}{2}\xi^2\right) \propto \xi^n \exp\left(-\frac{1}{2}\xi^2\right) \rightarrow 0, \quad \xi \rightarrow \pm\infty.$$

Hermite polynomials

$$C_{k+2} = \frac{2k - (\lambda - 1)}{(k+2)(k+1)} C_k = -\frac{2(n-k)}{(k+2)(k+1)} C_k.$$

$$C_k = -\frac{(k+2)(k+1)}{2(n-k)} C_{k+2}.$$

$$m = n - k,$$

$$k = n - m,$$

$$C_{n-m} = -\frac{(n-m+2)(n-m+1)}{2m} C_{n-m+2}.$$

Hermite polynomials

$$C_{n-m} = -\frac{(n-m+2)(n-m+1)}{2m}C_{n-m+2}.$$

$$C_n = 2^n.$$

$$C_{n-2} = -n(n-1) \cdot 2^{n-2},$$

$$C_{n-4} = +\frac{n(n-1)(n-2)(n-3)}{2!} \cdot 2^{n-4},$$

$$C_{n-6} = -\frac{n(n-1)(n-2)(n-3)(n-4)(n-5)}{3!} \cdot 2^{n-6},$$

.....

$$H_n(z) = C_0 + C_2 z^2 + \cdots + 2^n z^n, \quad n = \text{even number},$$

$$H_n(z) = C_1 z + C_3 z^3 + \cdots + 2^n z^n, \quad n = \text{odd number}.$$

$$H_0(z) = 1,$$

$$H_1(z) = 2z,$$

$$H_2(z) = 4z^2 - 2,$$

$$H_3(z) = 8z^3 - 12z,$$

.....

Solutions to Schrödinger equation

$$\lambda = \frac{E}{\frac{1}{2}\hbar\omega} \Rightarrow E = \frac{1}{2}\lambda\hbar\omega,$$

$$E_n = \frac{1}{2}(2n + 1)\hbar\omega$$

$$E_n = n\hbar\omega + \frac{1}{2}\hbar\omega.$$

$$\psi_n(\xi) = C_n H_n(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right).$$

Generating function

Another definition of $H_n(z)$:

$$\exp(-s^2 + 2zs) = \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n.$$

$$\begin{aligned} H_n(z) &= \left. \frac{d^n}{ds^n} \exp(-s^2 + 2zs) \right|_{s=0} \\ &= \exp(z^2) \left. \frac{d^n}{ds^n} \exp(-(s-z)^2) \right|_{s=0} \\ &= (-)^n \exp(z^2) \left. \frac{d^n}{dz^n} \exp(-(s-z)^2) \right|_{s=0} \\ &= (-)^n \exp(z^2) \frac{d^n}{dz^n} \exp(-z^2), \end{aligned}$$

Generating function

$$H_n(z) = (-1)^n \exp(z^2) \frac{d^n}{dz^n} \exp(-z^2).$$

$$H_0(z) = 1,$$

$$H_1(z) = 2z,$$

$$H_2(z) = 4z^2 - 2,$$

$$H_3(z) = 8z^3 - 12z,$$

.....

Orthogonality

$$\exp(-t^2 + 2zt) = \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} t^n,$$

$$\exp(-s^2 + 2zs) = \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n,$$

$$\exp(-s^2 + 2zs) \cdot \exp(-t^2 + 2zt) = \sum_{m,n=0}^{+\infty} \frac{H_m(z)}{m!} \frac{H_n(z)}{n!} t^m s^n.$$

$$\begin{aligned} & \exp(-s^2 + 2zs) \cdot \exp(-t^2 + 2zt) \\ &= \exp(-t^2 + 2zt - s^2 + 2zs) \\ &= \exp(2ts + z^2) \cdot \exp(-(t + s - z)^2) \end{aligned}$$

Orthogonality

$$\exp(2ts + z^2) \cdot \exp(-(t + s - z)^2) = \sum_{m,n=0}^{+\infty} \frac{H_m(z)}{m!} \frac{H_n(z)}{n!} t^m s^n,$$

$$\exp(2ts) \cdot \exp(-(t + s - z)^2) = \sum_{m,n=0}^{+\infty} \frac{H_m(z)}{m!} \frac{H_n(z)}{n!} t^m s^n \exp(-z^2).$$

$$\begin{aligned} & \exp(2ts) \int_{-\infty}^{+\infty} \exp(-(t + s - z)^2) dz \\ &= \int_{-\infty}^{+\infty} \sum_{m,n=0}^{+\infty} \frac{H_m(z)}{m!} \frac{H_n(z)}{n!} t^m s^n \exp(-z^2) dz \\ &= \sum_{m,n=0}^{+\infty} \frac{1}{m!} \frac{1}{n!} t^m s^n \int_{-\infty}^{+\infty} H_m(z) H_n(z) \exp(-z^2) dz. \end{aligned}$$

Orthogonality

$$\begin{aligned}& \exp(2ts) \int_{-\infty}^{+\infty} \cdot \exp(-(t+s-z)^2) dz \\&= \exp(2ts) \int_{-\infty}^{+\infty} \exp(-y^2) dy \\&= \exp(2ts) \sqrt{\pi} \\&= \sqrt{\pi} \sum_{n=0}^{+\infty} \frac{1}{n!} (2st)^n, \\&= \sum_{n=0}^{+\infty} \frac{\sqrt{\pi} 2^n}{n!} t^n s^n,\end{aligned}$$

Orthogonality

$$\sum_{m,n=0}^{+\infty} \frac{1}{m!} \frac{1}{n!} t^m s^n \int_{-\infty}^{+\infty} H_m(z) H_n(z) \exp(-z^2) dz = \sum_{n=0}^{+\infty} \frac{\sqrt{\pi} 2^n}{n!} t^n s^n,$$

$$\int_{-\infty}^{+\infty} H_m(z) H_n(z) \exp(-z^2) dz = \sqrt{\pi} 2^n n! \delta_{mn}.$$

$$\exp\left(-s^2 + 2zs\right) = \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n,$$

$$\frac{d}{ds} \exp\left(-s^2 + 2zs\right) = \frac{d}{ds} \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n,$$

$$\frac{d}{ds} \exp\left(-s^2 + 2zs\right)$$

$$= (-2s + 2z) \cdot \exp\left(-s^2 + 2zs\right) = (-2s + 2z) \cdot \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n$$

$$= -2 \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^{n+1} + 2z \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n$$

$$= -2 \sum_{n=1}^{+\infty} \frac{H_{n-1}(z)}{(n-1)!} s^n + 2z \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n,$$

$$\begin{aligned}\frac{d}{ds} \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n &= \sum_{n=1}^{+\infty} \frac{H_n(z)}{n!} n s^{n-1} = \sum_{n=0}^{+\infty} \frac{H_{n+1}(z)}{n!} s^n, \\ -2 \sum_{n=1}^{+\infty} \frac{H_{n-1}(z)}{(n-1)!} s^n + 2z \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n &= \sum_{n=0}^{+\infty} \frac{H_{n+1}(z)}{n!} s^n.\end{aligned}$$

For the term of s^0 ,

$$2z H_0(z) = H_1(z) = 2z.$$

For the term of s^n ($n > 0$),

$$\begin{aligned}-2 \frac{H_{n-1}(z)}{(n-1)!} + 2z \frac{H_n(z)}{n!} &= \frac{H_{n+1}(z)}{n!}, \\ -2n H_{n-1}(z) + 2z H_n(z) &= H_{n+1}(z),\end{aligned}$$

$$H_{n+1}(z) - 2z H_n(z) + 2n H_{n-1}(z) = 0.$$

Recursive formula

For $n = 0$,

$$\begin{aligned} & H_{n+1}(z) - 2z H_n(z) + 2n H_{n-1}(z) \\ = & H_1(z) - 2z H_0(z) = 2z - 2z \\ = & 0, \end{aligned}$$

$$H_{n+1}(z) - 2z H_n(z) + 2n H_{n-1}(z) = 0.$$

Therefore,

$$H_{n+1}(z) - 2z H_n(z) + 2n H_{n-1}(z) = 0. \quad 0 \leq n \in \mathbb{Z}.$$

$$\exp\left(-s^2 + 2zs\right) = \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n,$$

$$\frac{d}{dz} \exp\left(-s^2 + 2zs\right) = \frac{d}{dz} \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n,$$

$$2s \cdot \exp\left(-s^2 + 2zs\right) = \sum_{n=0}^{+\infty} \frac{H'_n(z)}{n!} s^n,$$

$$2s \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^n = \sum_{n=0}^{+\infty} \frac{H'_n(z)}{n!} s^n,$$

$$2 \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^{n+1} = \sum_{n=0}^{+\infty} \frac{H'_n(z)}{n!} s^n,$$

$$2 \sum_{n=0}^{+\infty} \frac{H_n(z)}{n!} s^{n+1} = \sum_{n=0}^{+\infty} \frac{H'_n(z)}{n!} s^n,$$

$$2 \sum_{n=1}^{+\infty} \frac{H_{n-1}(z)}{(n-1)!} s^n = \sum_{n=0}^{+\infty} \frac{H'_n(z)}{n!} s^n.$$

For the term of s^0 ,

$$0 = H'_1(z) = 0.$$

For the term of s^n ($n > 0$),

$$2 \frac{H_{n-1}(z)}{(n-1)!} = \frac{H'_n(z)}{n!},$$

$$H'_n(z) = 2n H_{n-1}(z).$$

Recursive formula

For $n = 0$,

$$H'_0(z) = 1' = 0 = 2 \times 0 \times H_{-1}(z),$$

$$H'_n(z) = 2n H_{n-1}(z).$$

Therefore,

$$H'_n(z) = 2n H_{n-1}(z), \quad 0 \leq n \in \mathbb{Z}.$$

Solution to Hermite equation

$$u'' - 2zu' + (\lambda - 1)u = 0, \quad \lambda = 2n + 1$$

$$H'_n(z) = 2n H_{n-1}(z),$$

$$\begin{aligned} H''_n(z) &= 2n H'_{n-1}(z) \\ &= 4n(n-1) H_{n-2}(z), \end{aligned}$$

$$\begin{aligned} &H''_n(z) - 2z H'_n(z) + (\lambda - 1) H_n(z) \\ = &H''_n(z) - 2z H'_n(z) + 2n H_n(z) \\ = &4n(n-1) H_{n-2}(z) - 2z \cdot 2n H_{n-1}(z) + 2n H_n(z) \\ = &2n [H_n(z) - 2z H_{n-1}(z) + 2(n-1) H_{n-2}(z)], \end{aligned}$$

$$\begin{aligned}
& H_n''(z) - 2z H_n'(z) + (\lambda - 1) H_n(z) \\
&= 2n [H_n(z) - 2z H_{n-1}(z) + 2(n-1) H_{n-2}(z)] \\
&= 0,
\end{aligned}$$

$$H_{n+1}(z) - 2z H_n(z) + 2n H_{n-1}(z) = 0. \quad 0 \leq n \in \mathbb{Z}.$$

Conclusion

Hermite polynomials $H_n(z)$ are solutions to Hermite equation,

$$u'' - 2zu' + (\lambda - 1)u = 0, \quad \lambda = 2n + 1.$$

Polynomial solutions to Hermite equation,

$$u'' - 2zu' + (\lambda - 1)u = 0, \quad \lambda = 2n + 1,$$

must satisfy the condition:

$$C_{n-m} = -\frac{(n-m+2)(n-m+1)}{2m}C_{n-m+2}.$$

When setting

$$C_n = 2^n,$$

the Polynomial solution for each n ($0 \leq n \in \mathbb{Z}$) is unique.

$$\begin{aligned}\frac{d}{dz} \exp(-z^2) &= (-2z) \exp(-z^2), \\ \frac{d^2}{dz^2} \exp(-z^2) &= (-2z)^2 \exp(-z^2) + \cdots, \\ &\dots\dots\dots,\end{aligned}$$

$$\frac{d^n}{dz^n} \exp(-z^2) = (-2z)^n \exp(-z^2) + \dots$$

$$\begin{aligned} H_n(z) &= (-1)^n \exp(z^2) \frac{d^n}{dz^n} \exp(-z^2) \\ &= (-1)^n \exp(z^2) \left[(-2z)^n \exp(-z^2) + \dots \right] \\ &= (-1)^n (-2z)^n + \dots \\ &= 2^n z^n + \dots \end{aligned}$$

$$C_n = 2^n.$$

Conclusion

The two definitions of Hermite polynomial are identical.

Orthonormalization

$$\psi_n(\xi) = C_n H_n(\xi) \cdot \exp\left(-\frac{1}{2}\xi^2\right), \quad \xi = \alpha x = \sqrt{\frac{m\omega}{\hbar}}x.$$

$$\psi_n(x) = \psi_n(\xi) = C_n H_n(\alpha x) \cdot \exp\left(-\frac{1}{2}(\alpha x)^2\right).$$

$$\begin{aligned} & \int_{-\infty}^{+\infty} \psi_m^*(x) \psi_n(x) dx \\ &= C_m^* C_n \int_{-\infty}^{+\infty} H_m(\alpha x) \exp\left(-\frac{1}{2}(\alpha x)^2\right) H_n(\alpha x) \exp\left(-\frac{1}{2}(\alpha x)^2\right) dx \\ &= C_m^* C_n \int_{-\infty}^{+\infty} H_m(\alpha x) H_n(\alpha x) \exp\left(-(\alpha x)^2\right) dx \\ &= \frac{C_m^* C_n}{\alpha} \int_{-\infty}^{+\infty} H_m(z) H_n(z) \exp\left(-z^2\right) dz. \end{aligned}$$

Orthonormalization

$$\int_{-\infty}^{+\infty} H_m(z) H_n(z) \exp(-z^2) dz = \sqrt{\pi} 2^n n! \delta_{mn},$$

$$\int_{-\infty}^{+\infty} \psi_m^*(x) \psi_n(x) dx = \frac{C_m^* C_n}{\alpha} \sqrt{\pi} 2^n n! \delta_{mn}.$$

$$C_n = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{\frac{1}{2}},$$

$$\psi_n(x) = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{\frac{1}{2}} H_n(\alpha x) \cdot \exp\left(-\frac{1}{2}(\alpha x)^2\right),$$

$$\int_{-\infty}^{+\infty} \psi_m^*(x) \psi_n(x) dx = \delta_{mn}.$$

Summary

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2.$$

$$E_n = n\hbar\omega + \frac{1}{2}\hbar\omega.$$

$$\psi_n(x) = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{\frac{1}{2}} H_n(\alpha x) \cdot \exp\left(-\frac{1}{2}(\alpha x)^2\right), \quad \alpha = \sqrt{\frac{m\omega}{\hbar}}.$$

$$\int_{-\infty}^{+\infty} \psi_m^*(x) \psi_n(x) dx = \delta_{mn}.$$